

Sorting and Searching Algorithms - Questions & Answers

1. Which sorting algorithm does not use recursion?

Answer: Bottom-up Merge Sort and Heap Sort

2. Fastest sorting algorithm for small arrays?

Answer: Insertion Sort

3. Method used in Merge Sort?

Answer: Merging

4. Best for nearly sorted array?

Answer: Insertion Sort

5. Where does Uniform Binary Search fail?

Answer: Complexity of code

6. Best data structure for Priority Queue?

Answer: Heap

7. Quick Sort uses which method?

Answer: Partitioning

8. Disadvantage of Linear Search?

Answer: Greater time complexity compared to other searching algorithms

9. If swap is costly, which sorting to prefer?

Answer: Selection Sort

10. Sorting algorithm with worst-case $O(n^2)$?

Answer: Bubble Sort (Quick Sort also in worst case)

11. Best data structure for recursion?

Answer: Stack

12. Where is Linear Search used?

Answer: When list is small and for single search in unordered list

13. Algorithm to find kth largest element?

Answer: Selection Algorithm (Quickselect)

14. Best and worst case of ordered Linear Search?

Answer: $O(1)$, $O(n)$

15. Worst case recurrence of QuickSort?

Answer: $T(n) = T(n-1) + O(n)$, Time Complexity = $O(n^2)$

16. Sorting k-sorted array?

Answer: Heap Sort with $O(n \log k)$

17. Pivot in array 2 5 1 7 9 12 11 10?

Answer: Pivot could be either 7 or 9

18. Advantage of Selection Sort?

Answer: Requires no additional storage space

19. Fastest searching algorithm?

Answer: Binary Search

20. Worst case of Linear Search?

Answer: $O(n)$